

Active Directory SOC Detection Lab

Technical Documentation

A self-built blue-team detection environment: Active Directory domain, centralized Splunk logging, Sysmon-instrumented endpoints, and a Kali Linux attacker. Adversary techniques executed with Atomic Red Team and detected end-to-end in Splunk, mapped to MITRE ATT&CK.

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Domain:	dwn.lab / 192.168.10.0/24
Platform:	VMware Workstation

1. Project Overview

This document records the design, construction, and validation of a functional **Security Operations Center (SOC) detection lab** built from scratch in VMware Workstation. The objective was to create a realistic environment where offensive techniques can be executed against domain-joined endpoints and then detected through a centralized logging pipeline — replicating the daily workflow of a blue-team analyst.

The lab demonstrates the complete **attack** → **log** → **detection** cycle:

- An attacker (Kali Linux) executes techniques against domain-joined endpoints.
- Endpoints generate Windows Security and Sysmon telemetry.
- A Splunk Universal Forwarder ships logs to a central Splunk indexer.
- Detections are built in Splunk and mapped to MITRE ATT&CK.

Personal context: this is my first independent cybersecurity lab, built after ~1.6 years of study in the Information Security program at ITLA. It consolidates networking, Active Directory, SIEM, and adversary-emulation skills into a single end-to-end project.

2. Lab Architecture

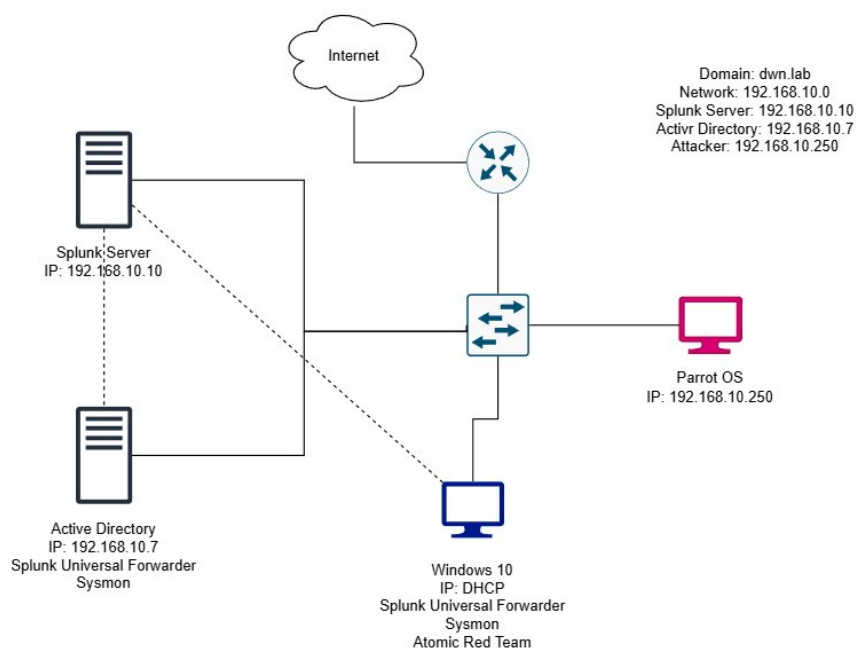


Figure 1 — Lab network topology (dwn.lab, 192.168.10.0/24).

Note: the diagram labels the attacker host generically; in this build the attacker machine is Kali Linux at 192.168.10.250.

Host	IP	Role
Splunk Server	192.168.10.10	Central indexer / search head
Active Directory	192.168.10.7	Domain Controller (dwn.lab), Splunk UF, Sysmon
Windows 10	192.168.10.100 (static)	Domain-joined endpoint; Splunk UF, Sysmon, Atomic Red Team

Host	IP	Role
Kali Linux	192.168.10.250	Attacker / offensive tooling

3. Components & Build

Splunk Server (Ubuntu Server)

Splunk Enterprise as the central indexer and search head. Receives forwarded data on port 9997; web UI on :8000 for searching and dashboards.

Active Directory (Windows Server)

Promoted to Domain Controller for dwn.lab. Sysmon installed for deep process/network telemetry; Splunk Universal Forwarder ships Security + Sysmon logs.

Windows 10 Endpoint

Joined to the dwn.lab domain. Sysmon + Splunk Universal Forwarder installed. Atomic Red Team for adversary emulation. Static IP for deterministic behavior.

Kali Linux Attacker

Primary offensive machine for the lab. Tooling used: nmap, hydra, crowbar.

4. Network Configuration

All VMs share a custom **VMnet in NAT mode** on 192.168.10.0/24, gateway at .1. A key design decision was to assign **static IPs outside the DHCP pool**. Early in the build the Windows 10 endpoint suffered intermittent connectivity caused by a short DHCP lease (~22 min) colliding with stale NAT/DHCP state on the host. Two changes made the lab deterministic:

- Assigned static IPs to the servers and the Windows 10 endpoint.
- Narrowed the VMware DHCP pool so it no longer overlapped the static addresses (.10, .100).

Setting	Value
IP (Win10)	192.168.10.100
Subnet Mask	255.255.255.0
Gateway	192.168.10.1
DNS	192.168.10.7 (AD / domain DNS)

Takeaway: in a detection lab, deterministic networking matters. A flapping endpoint IP breaks Universal Forwarder connectivity mid-exercise and wastes time chasing phantom Splunk problems.

5. Detection Pipeline

Endpoints (Windows 10 and the Domain Controller) generate Windows Security events and Sysmon telemetry. A Splunk Universal Forwarder on each endpoint ships logs to the indexer at 192.168.10.10:9997, where searches turn raw events into ATT&CK-mapped detections.

Endpoints (Win10 / AD) → Windows Security + Sysmon → Universal Forwarder → Splunk Indexer
→ Searches + ATT&CK

Telemetry sources

- Windows Security Event Log — authentication and account management (4624, 4625, 4720, 4732).
- Sysmon — process creation (Event ID 1), network connections, and granular host activity.

6. Attack Simulation & Detection

Two MITRE ATT&CK techniques were executed end-to-end and verified as detectable in the pipeline.

6.1 T1110 — Brute Force (RDP)

```
# Confirm RDP is listening
nmap -p 3389 192.168.10.100

# Brute force RDP with a password list
hydra -l jdepaula -P passwords.txt rdp://192.168.10.100
```

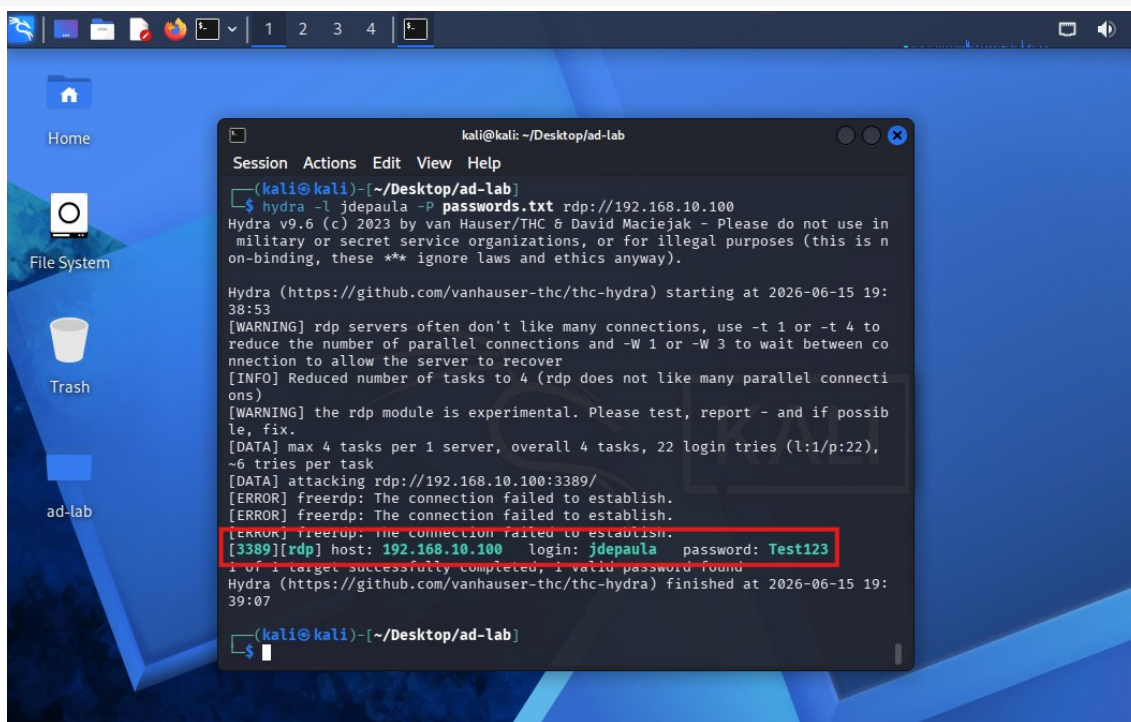


Figure 2 — hydra recovers the valid RDP credential (jdepaula / Test123).

Tooling note: crowbar was tried first but failed against modern Windows 10 RDP — its aging xfreerdp wrapper breaks the NLA handshake and reports every credential as invalid even when one is correct. hydra handled the handshake correctly and succeeded. **Lesson:** when an RDP brute-force tool reports universal failure, suspect the tool / NLA before the target configuration.

Event ID	Meaning
4625	Failed logon — a burst, one per wrong password
4624 (Logon Type 10)	Successful RemoteInteractive (RDP) logon

Detection pattern: many 4625 events from a single source IP (192.168.10.250), followed by a 4624 with Logon Type 10 from the same source.

```
index="endpoint" EventCode=4624 kali
```

The screenshot shows the Splunk web interface with the search query `index="endpoint" EventCode=4624 kali`. The search results show 1 of 27 events matched. The event details are displayed in a table format. The event occurred on 6/15/26 at 11:47:37.941 PM. The event details include:

- Nombre de estación de trabajo: kali (highlighted with a red arrow)
- Dirección de red de origen: 192.168.10.250
- Puerto de origen: 0
- host = target-pc
- source = WinEventLog:Security
- sourcetype = WinEventLog:Security

The interface also shows a list of fields on the left, including `host`, `source`, `sourcetype`, `ComputerName`, `Cuenta_virtual`, `Dirección de red de origen`, `Dominio_de_cuenta`, `Dominio_de_cuenta_de_red`, `EventCode`, `EventType`, `GUID de inicio de sesión`, `Id. de inicio de sesión`, and `Id_de_proceso`.

Figure 3 — Splunk shows the successful RDP logon (4624) sourced from the Kali attacker at 192.168.10.250.

6.2 T1136.001 — Create Account: Local Account

Invoke-AtomicTest T1136.001

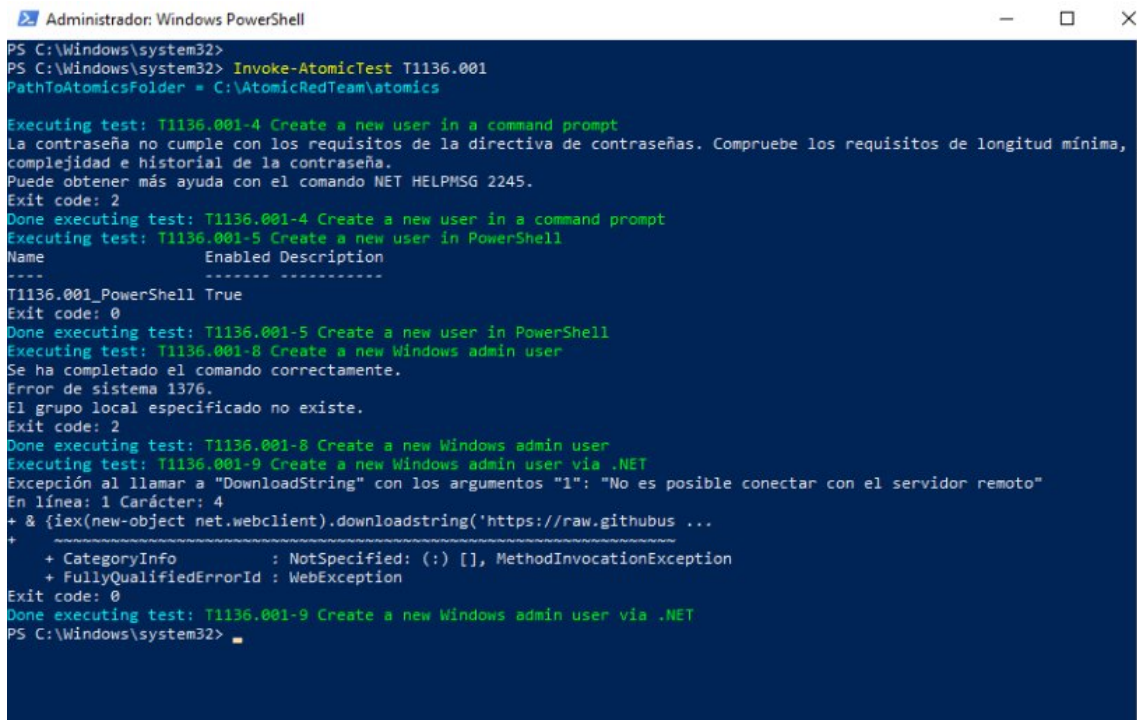


Figure 4 — Atomic Red Team executing the T1136.001 sub-tests on the endpoint.

Test	Result	Note
-4 (cmd)	Blocked	Password failed the domain password policy (complexity). The GPO worked.
-5 (PowerShell)	Success	New-LocalUser created the account.
-8 (admin user)	Partial	User created; group add failed — Spanish Windows uses 'Administradores'.
-9 (.NET)	Blocked	Outbound fetch to raw.githubusercontent.com failed (network/DNS).

Two of these tests created real accounts, generating exactly the telemetry a SOC should catch.

Event ID	Meaning
4720	A user account was created
4732	A member was added to a security-enabled local group
Sysmon 1	Process creation — net user / New-LocalUser execution

index="endpoint" New-LocalUser powershell

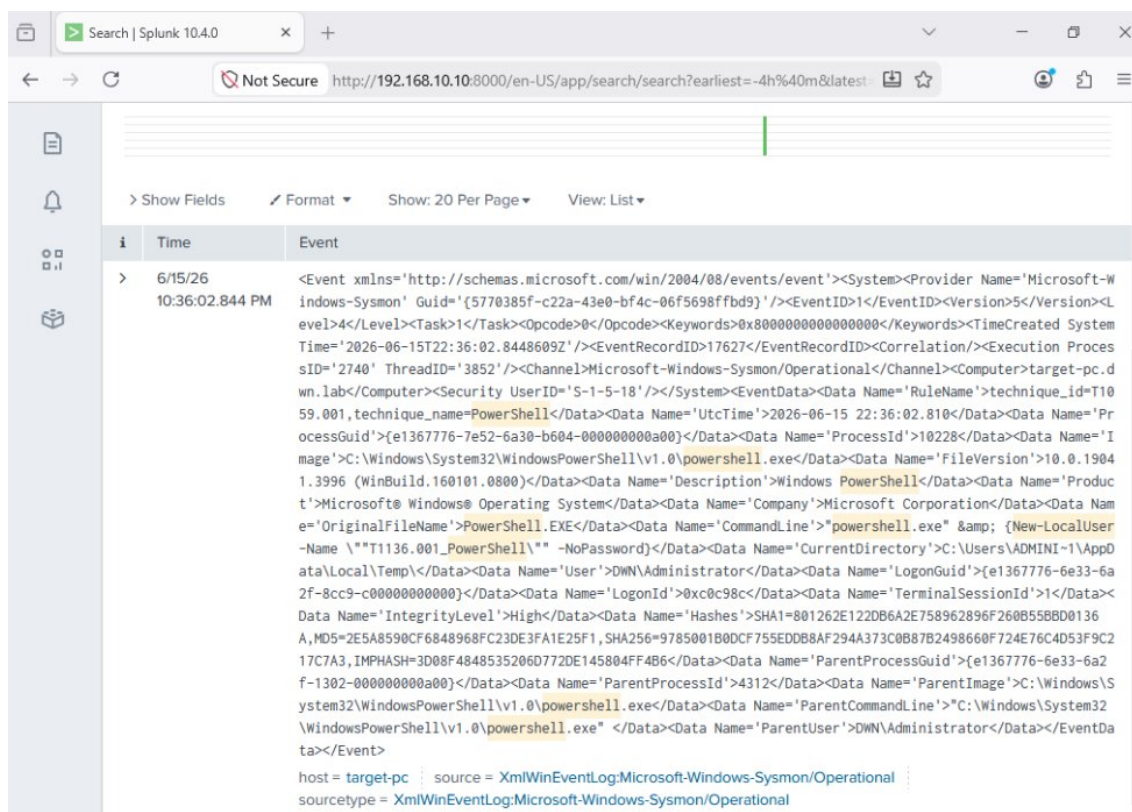


Figure 5 — Sysmon Event ID 1 in Splunk capturing the New-LocalUser process spawned by the atomic test.

7. Troubleshooting Log

Real problems encountered and resolved during the build — included because troubleshooting *is* the skill.

#	Problem	Root Cause	Fix
1	Win10 no internet in NAT (Linux VM failed)	Stale host NAT/DHCP state + short lease	Static IP outside pool; restart VMware NAT/DHCP services
2	Connectivity dropped every ~22 min	DHCP lease renewal vs. stale NAT state	Static IP killed the renewal cycle
3	Splunk halted searches	Disk below 5 GB free on /opt/splunk	Expanded virtual disk (3-stage LVM grow)
4	Expanded disk not visible in OS	Filesystem not grown after disk resize	growpart -> lvextend -> resize2fs
5	Invisible cursor in Linux VM	Graphics driver / virtual-hardware mismatch	Rebased VM Hardware Compatibility version
6	crowbar RDP brute force failed	Aging xfreerdp wrapper vs. Win10 NLA	Switched to hydra

7.1 Disk Expansion (LVM) — the 3-stage grow

A frequent gotcha: expanding the virtual disk in VMware does **not** grow the filesystem. Three layers must be extended in order.

```
# 1. Grow the partition to claim the new disk space
sudo growpart /dev/sda 3

# 2. Extend the LVM logical volume into the freed space
sudo lvextend -l +100%FREE /dev/ubuntu-vg/ubuntu-lv

# 3. Grow the ext4 filesystem on top
sudo resize2fs /dev/ubuntu-vg/ubuntu-lv

# Verify
df -h /opt/splunk/
```

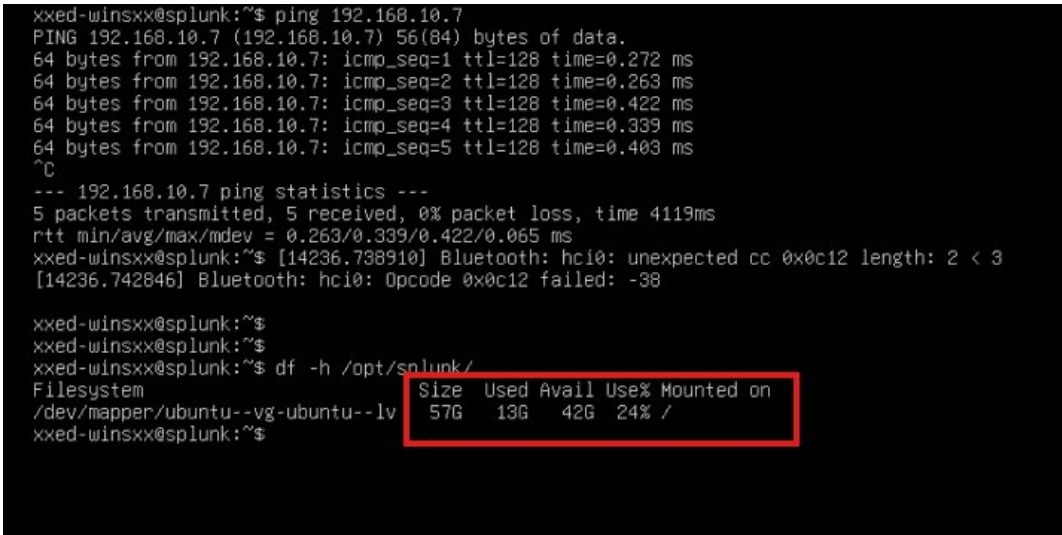


Figure 6 — df -h confirms the filesystem grew to 57G after the LVM expansion.

8. Lessons Learned

Deterministic infrastructure first. Static IPs and a clean DHCP scope eliminated an entire class of intermittent failures that masqueraded as Splunk problems.

The OS sits below the application. A “Splunk stopped searching” error was really a full disk — always check the layers underneath before debugging the app.

Tools are not interchangeable. crowbar and hydra target the same protocol but handle the RDP/NLA handshake differently. When a tool reports universal failure, suspect the tool first.

Localization matters in offensive tooling. Atomic tests hardcoded for 'Administrators' fail silently on Spanish-language Windows ('Administradores').

Troubleshooting is the deliverable. A lab that built perfectly on the first try teaches less than one fought into existence.

9. MITRE ATT&CK Coverage

Technique	ID	Tactic	Status
Brute Force	T1110	Credential Access	Detected
Create Account: Local Account	T1136.001	Persistence	Detected

Next Steps

- Build saved searches / alerts in Splunk for T1110 and T1136.001.
- Expand Atomic Red Team coverage (persistence, lateral movement, exfiltration).
- Add dashboards mapping detections to the MITRE ATT&CK matrix.
- Tune index retention (frozenTimePeriodInSecs) to keep disk usage bounded.
- Snapshot each VM in a known-good state for fast lab recovery.

Built as a hands-on blue-team / detection engineering project — demonstrating SOC pipeline design, adversary emulation, and ATT&CK-mapped detection.

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